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RESEARCH ARTICLE

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Elucidating the molecular role of MUC5B in progressive lung adenocarcinoma: Prospects for early diagnosis

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Abstract

Gel-forming mucin MUC5B is significantly deregulated in lung adenocarcinoma (LUAD), however, its role in tumor progression is not yet clearly understood. Here, we used an integrated computational-pipeline-initiated with gene expression analysis followed by network, functional-enrichment, O-linked glycosylation analyses, mutational profiling, and immune cell infiltration estimation to functionally characterize MUC5B gene in LUAD. Thereafter, clinical biomarker validation was supported by the overall survival (OA) and comparative expression profiling across clinical stages using computational algorithms. The gene expression profile of LUAD identified MUC5B to be significantly up-regulated (logFC: 2.36; *p*-value: 0.01). Network analysis on LUAD interactome screened MUC5B-related genes, having key enrichment in immune suppression and O-linked glycosylation with serine–threonine-rich tandem repeats being highly glycosylated. Furthermore, positive correlation of mutant MUC5B with immune cells in tumor microenvironment (TME) such as cancer-associated fibroblasts and myeloid-derived suppressor cells indicates TME-mediated tumor progression. The positive correlation with immune inhibitors suggested the enhanced tumor proliferation mediated by MUC5B. Structural stability due to genetic alterations identified overall rigid N–H-backbone dynamics (S^2: 0.756), indicating an overall stable mutant protein. Moreover, the low median OA (<50 months) with a hazard ratio of 1.4 and clinical profile of MUC5B gene showed high median expression corresponding to lymph node (N2) and tumor (T3) stages. Our study concludes by highlighting the functional role of O-glycosylated and mutant MUC5B in promoting LUAD by immune suppression. Further, clinical gene expression validation of MUC5B suggests its potential role as a diagnostic biomarker for LUAD metastasis.

KEYWORDS
immune infiltration, lung cancer, mucins, mutations, O-linked glycosylation, tumor microenvironment

1 | INTRODUCTION

Lung cancer being the leading cause of cancer deaths and cases reported asserts the vulnerable survival outcomes associated with the tumor. With an estimate of 2.2 million new cases (11.4%) and 1.7

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million cancer deaths (18%), lung cancer still remains a disincentive that requires more perception.[1] Non-small cell lung cancer (NSCLC) accounts for more than 80% of lung cancers reported and among them, lung adenocarcinoma (LUAD) is the most common pathological subtype of lung cancer. While the current medical intervention is focused on EGFR and KRAS inhibitors, the need for a more targeted biomarker is necessary to prevent LUAD progression. Moreover, the incidence of LUAD is not restricted to smokers, and its presence equitably in non-smokers further makes the tumor more dangerous. Additionally, the immune evasion and resistance are posing a huge blockade to the existing therapeutic regimens, thereby pushing for treatment strategies based on immunotherapy.[2]

The high molecular weight glycoprotein, mucins covering the epithelial surfaces of mammalian respiratory, reproductive, and gastrointestinal tracts have long been reported to aid in carcinogenesis. About 11 mucins are expressed in the lungs and are majorly categorized into transmembrane and secretory. The former plays a role in lubricating and protecting the tissues from infectious diseases, while the latter are gel-forming mucins creating polymeric mucins that adsorb water to form mucus. Among the secreted mucins, MUC5B and MUC5AC are predominantly found in the proximal and distal airways.[3] Furthermore, the mucin-enriched O-linked glycosylation dictates and mediates the cell–cell adhesion, immune recognition, and activation of oncogenic signaling pathways.[4] Deregulated or aberrant glycosylation promotes tumor proliferation by regulating invasion and metastasis by the cross-talk between the tumor microenvironment (TME) and glycans.[5] Few reports also suggested the cross-talk among mucin-associated genes and receptor tyrosine kinases (RTKs) through the activation of signaling pathways and glycosylation.[6–8] Reports on the mucin's ability to mimic tumor cell surface and exhibiting oncogenic microenvironment have also been evaluated. The mucus formed from the mucins contributes to tumor growth by capturing cytokines and growth factors. Among the gel-forming mucins, MUC5B has been reported to be promoting respiratory disorders such as idiopathic pulmonary fibrosis (IPF) and cystic fibrosis (CF). Additionally, the potential role of MUC5B in promoting chemoresistance in breast cancer has been reported.[9] Moreover, the crosstalk of mucins such as MUC5B in chronic lung diseases are also reported.[10] The mutational association of MUC5B across pan-cancer and their aberrant expression has been quantified.[11,12] Even though adenocarcinoma of the lung usually evolves around the mucous gland and represents about 40% of all lung cancer, the role of MUC5B in LUAD is still not clearly understood. Therefore, the present study focuses on the role of MUC5B gene in LUAD.

The timely and accurate detection of LUAD requires the specific identification of tumor cell biomarkers, hence in our present study, we applied an integrated in silico approach[13–21] to comprehensively understand the functional significance of MUC5B in LUAD by gene expression profiling, clustering-functional enrichment analyses, genetic alterations, tumor-immune cell infiltration and glycosylation patterns. Further, the potentiality of MUC5B as theranostic biomarker was validated using clinical profile and survival analyses.

2 | MATERIALS AND METHODS

2.1 | Data collection and differential gene expression

Microarray expression profiles of LUAD samples were retrieved from GEO database with accession number GSE5364 consisting of 18 lung tumor samples and 12 lung normal samples obtained using expression profiling by array. The expression profiling was performed using Affymetrix Human Genome U133A Array (HG-U133A). We mined unbiased experimental leads and associated molecular evidence. The dataset chosen as a lead is referred from an experimental setup, which focused on solid tumor-specific protein markers cross-referenced from available clinical evidence (341 samples). The suggested protein markers were not patient-specific and rather demographically uniform. The same was further validated experimentally.[22] We expanded the available lead by integrating with the expression profiling from The Cancer Genome Atlas (TCGA) and Genotype-Tissue Expression Database (GTEx) consortia comprising 585 cases with more than 10,000 samples of LUAD patients spanning across four repositories.[23–26] The expanded dataset will validate the expression profile of MUC5B, further supporting its role as diagnostic biomarker for LUAD patients. The resulting samples were then grouped into tumor and normal samples for differential gene expression using GEO2R. The differentially expressed genes were obtained using Benjamini & Hochberg with no force normalization and with a statistical significant p-value <0.05.[27]

2.2 | Interactome construction and analyses

The interaction data of the significantly up-regulated genes was then curated using the STRING database with a medium confidence score (CS) of >0.49. The medium CS depicts the statistically significant experimentally proven and predicted interactions among the nodes. Furthermore, the evidences include those that have been curated from text mining, experiments, databases, co-expression, neighborhood, gene fusion, and co-occurrence.[28] Thereafter, LUAD interactome was constructed using Cytoscape, which was used to further measure the topological metrics and interconnected clusters using cytoHubba and MCODE respectively. Clustering analysis using molecular complex detection algorithm analyzed the interactome with a degree cut-off of two and k-core parameter of two using MCODE plugin. Clustering was performed using a three-step process starting from vertex weighting, molecular complex prediction, and post-processing. Clustering analysis aims to discover the probable functionally related gene clusters that interact among themselves to perform a specific biological purpose.[29]

2.3 | Functional enrichment analysis

An automated pathway-enrichment analysis was performed on the MUC5B-present cluster using hypergeometric testing by the STRING

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database. The database performed overrepresentation tests for gene ontology (GO) terms such as biological process (BP), cellular components (CC), enriched tissues, and reactome pathways. Statistically significant enriched terms with a p-value <0.05 were considered along with the overlapping genes against a statistical background of the entire genome.

2.4 | O-linked glycosylation pattern prediction

Glycosylation Predictor utilizes the random forest machine learning algorithm based on decision trees and trepan algorithm designed to comprehend neural networks for predicting glycosylation sites. The server acquires glycosylation sites from OGLYCBASE and analyzes the amino acid frequencies, secondary structure, and surface accessibility to screen the probable glycosylation sites.[30] The O-linked glycosylation sites predicted were thereafter cross-validated with the serine/threonine tandem repeats found in mucin using ProtParam server.[31]

2.5 | O-linked glycan prediction

GlyGen database was used to predict the MUC5B-associated glycans, which is an open-source web-interface comprising of glycoinformatics data integrating tools to explore glycoscience data. The GlyGen portal currently consists of N- and O- glycans across human, rat, and mouse species. The portal has multiple queries namely glycan, protein, GlyGen mapper, composition, structure, site, and motif search for identifying the diverse glycans and associated glycoproteins.[32]

2.6 | Transcription factor-miRNA network

miRNet web-interface provides an easy-to-use platform for the elucidation of microRNA and their target genes. The latest version has upgraded and incorporated miRNAs-target genes, their transcriptional factors, and probable SNPs that can alter miRNA expression, their binding, and the disease associated with it. In our study, the miRNA and the transcription factors associated with the target gene were analyzed.[33,34]

2.7 | Immune cell infiltration estimation

Tumor Immune Estimation Resource (TIMER 2.0) comprehensively analyzes the immune infiltrates across diverse cancer types. TIMER 2.0 estimates tumor immune infiltrate population against transcriptomic changes in diverse tumors and are investigated across four different modules for the correlation of immune cells and clinical features. The mutation module considers all the non-synonymous somatic mutations associated with the target genes for the particular TCGA cancer type. The association is categorized into three namely, higher levels in mutant (p < 0.05; log2FC > 0), lower levels in mutant (p < 0.05; log2FC < 0), and not significant (p > 0.05). Here, we identified the probable association between mutant MUC5B and immune cells in LUAD.[35] Thereafter, the correlation of MUC5B gene and immunomodulators was analyzed using TISIDB, an integrated repository for screening tumor-immune systems interactions. Here, we analyze the correlation of target gene expression with immunomodulators (stimulators and inhibitors) using Spearman correlation and Wilcoxon test.[36]

2.8 | Mutational landscape of MUC5B functional domains

The genetic alterations associated with MUC5B were analyzed using cBioportal, a web resource for multi-dimensional analysis and visualization of cancer genomics data. Here, the genetic alterations associated with the target gene pertaining to LUAD were analyzed.[37] Further, single nucleotide polymorphisms (SNPs) associated with MUC5B were retrieved from BioMuta for lung cancer against their functional domains identified from InterPro database.[38,39] Thereafter, the functional impact of the SNPs was identified using PredictSNP. The open-source web server is extensively used for the evaluation of SNPs and their effects on protein function. The tool has constructed independent datasets with all duplicates removed and compared across eight established tools such as MAPP, PANTHER, PolyPhen-1, SIFT, nsSNPAnalyzer, PhD-SNP, PolyPhen-2, and SNAP. The CSs of these integrated tools were computed and observed with the PredictSNP CS.[40] Thereafter, the stabilizing and/or destabilizing mutations and backbone dynamics of the parent and mutant MUC5B were predicted using iMutant and sequence-based DynaMine. The server predicts the residue-level potential of protein in the form of the N–H S^2 order parameter, where the S^2 is predicted based on the NMR chemical shift value. The server computes S^2 pred value ranging from zero to one, with one indicating fully restricted conformation and zero indicating highly dynamic conformation.[41,42]

2.9 | Gene expression profiling of other mucins in LUAD

The gene expression profile of mucins in LUAD was identified using various gene expression databases like TNM plot,[23] GEPIA,[24] BioXpress,[38] and oncoDB database.[26] The databases curate data from both RNA-seq and microarray techniques into global repositories such as Genomic Data Commons (GDC), TCGA, and GTEx.

2.10 | Cancer biomarker validation

muTarget is a web-based tool for cancer biomarker or target discovery based on genomic and transcriptomic data across 18 different types of tumors. The web server identifies changes in gene expression

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pertaining to gene mutations as well as identifies mutations resulting in the alteration of gene expression. The server can be applied in two ways namely, Genotype, which identifies genes showing altered expression as a result of mutated input gene, and Target, which identifies the mutations that result in altered expression of input gene. The analysis was conducted on a sample size of 508 LUAD samples and the specific parameters taken into consideration include all somatic mutations with default threshold (fold change of 1.44) and *p*-value of <0.01. Genes with mean expression <100 were excluded and statistically significant results were generated.[43]

2.11 | Overall survival analysis

Overall survival (OA) statistically depicting the clinical outcome with low or high expression of cancer-related genes is analyzed using GEPIA. The tool provides the hazard ratio (HR), which is the probability of an event to occur when compared across two distinct groups calculated using log-rank statistics with a significant *p*-value. The web server integrates gene expression, and corresponding clinical data from TCGA and GTEx studies to characterize the differentially expressed genes across various cancer cases and their clinical significance. Further, we also checked the difference in the median survival (MS) of LUAD patients among various tumor stages using Km plotter. The MS translates the survival outcomes among the high and low expression groups, indicating the impact of the expression of gene on the survival of the patients.[44]

2.12 | Clinical profiling

Clinical parameter analysis was performed using OncoDB, an online database that explores altered gene expression of various TCGA cancer studies with clinical parameters such as pathological metastatic (M), lymph node (N), tumor, race, gender, and smoking. The database is also an online repository of differential methylation patterns and gene expression integrated through RNA-Seq from over 10,000 cancer samples from TCGA. The statistical analysis for gene expression levels in cancer versus normal is performed using normalized RNA seq data expressed in terms of transcript per million (TPM). The results were further supported by the TCGA RNA seq data using UALCAN database. The expression pattern of MUC5B gene was investigated across tumor stages and lymph metastasis stages.[45] A comprehensive flowchart representing the workflow of the interactome analyses is provided in Figure 1.

3 | RESULTS

3.1 | Screening differentially expressed genes and construction of LUAD interactome

The gene expression analysis of LUAD samples resulted in 671 up-regulated genes with a logFC value ≥1 and 855 down-regulated genes with a logFC value of ≤−1 with a *p*-value threshold of <0.05. Thereafter,

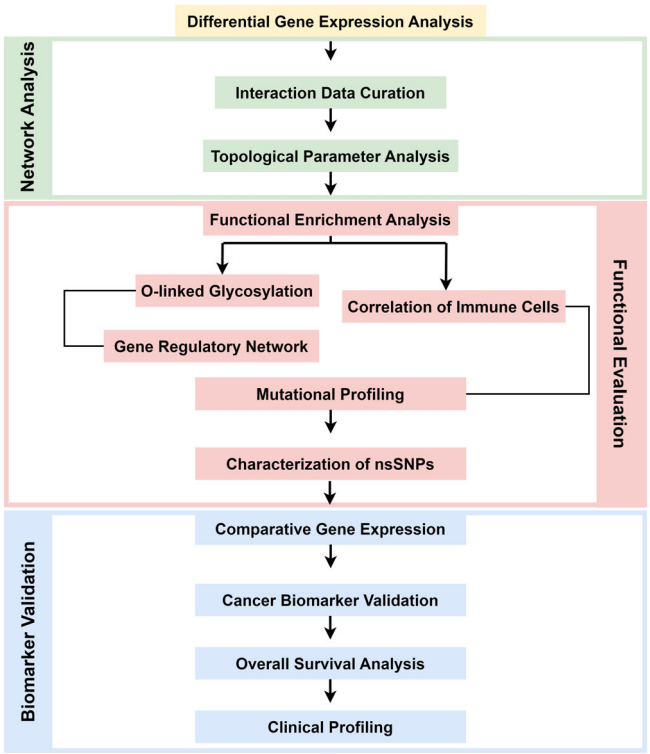


FIGURE 1 A systematic workflow depicting the interactive analyses employed for the elucidation of the molecular role of MUC5B.

the LUAD interactome was constructed using 659 up-regulated genes and 8321 edges with an average node degree of 25.3 (Table S1). The *MUC5B* gene was significantly up-regulated with a logFC value of 2.366 and a *p*-value of 0.01 as depicted in the volcano plot (Figure 2A).

3.2 | Topological parameter and clustering analyses

Network analysis computed topological metrics for each node in LUAD interactome and *MUC5B* was identified with a degree of 18, betweenness centrality of 2277.12, closeness centrality of 215.1, and clustering coefficient of 0.26 (Table S2). Clustering analysis on the LUAD interactome revealed 18 highly interconnected clusters with *MUC5B* gene present in cluster C4 with 8 nodes and 48 edges having a MCODE score of 6.857. The 18 clusters identified from the LUAD interactome are provided in Table S3. The interconnected nodes in cluster C4 containing MUC5B were *GCNT3*, *MUC5AC*, *B3GNT3*, *GALNT14*, *MUC3A*, *MUC16*, and *GALNT2* as depicted in Figure 2B.

3.3 | Functional enrichment on MUC5B-related cluster

The functional enrichment of the MUC5B-related cluster genes namely *GCNT3*, *MUC5AC*, *B3GNT3*, *GALNT14*, *MUC3A*, *MUC16*, and *GALNT2*

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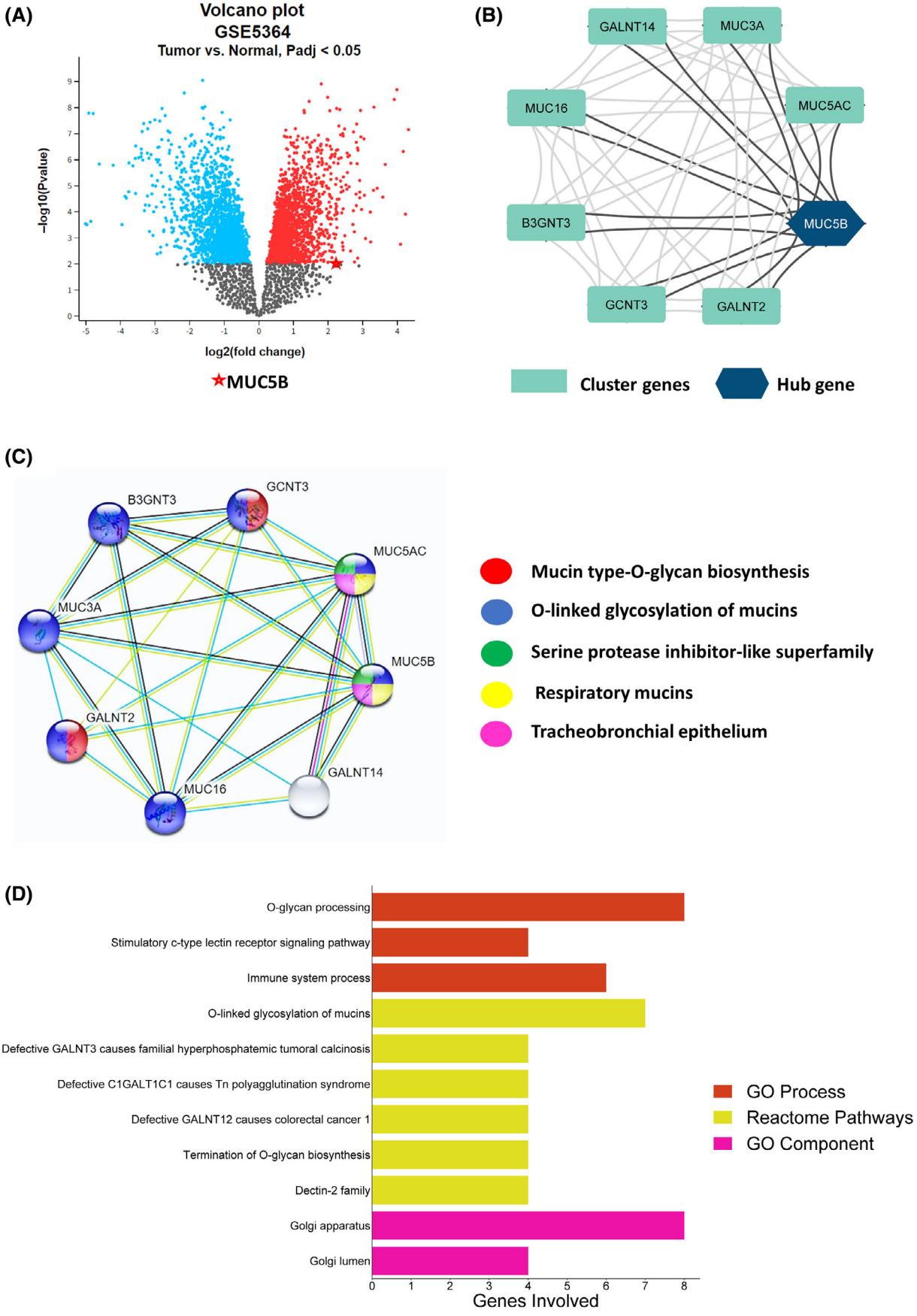


FIGURE 2 Gene expression, clustering, and functional enrichment analyses (A) Volcano plot depicting significantly up-regulated MUC5B (B) MUC5B-related cluster genes screened using MCODE algorithm (C) Functional role of MUC5B-related cluster genes (D) Bar plot depicting enriched reactome pathways, cellular components and biological processes of MUC5B-related cluster genes. GO, gene ontology.

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(Figure 2C,D) showed enrichment of O-glycan processing (GO:0016266), Stimulatory c-type lectin receptor signaling pathway (GO:0002223) and the Immune systems process (GO:0002376) as the major BPs. Golgi apparatus (GO:005794) and Golgi lumen (GO:0005796) were enriched CCs. O-linked glycosylation of mucins (HSA-913709), Defective GALNT3 caused familial hyperphoshatemic tumoral calcinosis (HFTC) (HSA-5083625), Defective GALNT12 causes colorectal cancer 1 (CRCS1) (HSA-5083636), Termination of O-glycan biosynthesis (HSA-977068) and Dectin-2 family (HSA-5621480) were enriched biological pathways. The enriched tissues of *MUC5B* were Gastrointestinal tract (BTO:0000511), Mucosa (BTO:0000886), Goblet cell (BTO:001540), Tracheobronchial epithelium (BTO:002925), Respiratory mucus (BTO:0003603) and submucosal gland (BTO:0003751) and the complete list of GO terms are summarized in Table 1.

TABLE 1 Functional enrichment analysis of MUC5B-present cluster identified biological process, cellular components, reactome pathways, tissues and compartments using STRING database.

Category	Term ID	Term description	Observed gene count	False discovery rate	Genes involved
Biological Process	GO:0016266	O-glycan processing	8	1.22E-16	*GALNT14*, *B3GNT3*, *GALNT2*, *MUC3A*, *GCNT3*, *MUC16*, *MUC5B*, *MUC5AC*
	GO:0002223	Stimulatory c-type lectin receptor signaling pathway	4	7.68E-05	*MUC3A*, *MUC16*, *MUC5B*, *MUC5AC*
	GO:0002376	Immune system process	6	0.0299	*GALNT2*, *MUC3A*, *GCNT3*, *MUC16*, *MUC5B*, *MUC5AC*
Cellular components	GO:0005794	Golgi apparatus	8	3.20E-06	*GALNT14*, *B3GNT3*, *GALNT2*, *MUC3A*, *GCNT3*, *MUC16*, *MUC5B*, *MUC5AC*
	GO:0005796	Golgi lumen	4	4.57E-05	*MUC3A*, *MUC16*, *MUC5B*, *MUC5AC*
Reactome	HSA-913709	O-linked glycosylation of mucins	7	6.91E-14	*B3GNT3*, *GALNT2*, *MUC3A*, *GCNT3*, *MUC16*, *MUC5B*, *MUC5AC*
	HSA-5083625	Defective GALNT3 causes familial hyperphosphatemic tumoral calcinosis (HFTC)	4	4.01E-08	*MUC3A*, *MUC16*, *MUC5B*, *MUC5AC*
	HSA-5083632	Defective C1GALT1C1 causes Tn polyagglutination syndrome (TNPS)	4	4.01E-08	*MUC3A*, *MUC16*, *MUC5B*, *MUC5AC*
	HSA-5083636	Defective GALNT12 causes colorectal cancer 1 (CRCS1)	4	4.01E-08	*MUC3A*, *MUC16*, *MUC5B*, *MUC5AC*
	HSA-977068	Termination of O-glycan biosynthesis	4	6.17E-08	*MUC3A*, *MUC16*, *MUC5B*, *MUC5AC*
	HSA-5621480	Dectin-2 family	4	9.69E-08	*MUC3A*, *MUC16*, *MUC5B*, *MUC5AC*
Tissues	BTO:0000511	Gastrointestinal tract	6	0.0017	*GALNT14*, *B3GNT3*, *MUC3A*, *GCNT3*, *MUC5B*, *MUC5AC*
	BTO:0000886	Mucosa	4	0.0017	*B3GNT3*, *MUC3A*, *MUC5B*, *MUC5AC*
	BTO:0001540	Goblet cell	2	0.0017	*MUC5B*, *MUC5AC*
	BTO:0002925	Tracheobronchial epithelium	2	0.0017	*MUC5B*, *MUC5AC*
	BTO:0003603	Respiratory mucus	2	0.0017	*MUC5B*, *MUC5AC*
	BTO:0003751	Submucosal gland	2	0.0017	*MUC5B*, *MUC5AC*
	BTO:0004524	Germinal epithelium	2	0.0017	*MUC16*, *MUC5AC*
	BTO:0001307	Stomach	3	0.0188	*GALNT14*, *B3GNT3*, *MUC5AC*
	BTO:0000634	Integument	5	0.0427	*B3GNT3*, *MUC3A*, *MUC16*, *MUC5B*, *MUC5AC*
	BTO:0001491	Viscus	7	0.0427	*GALNT14*, *B3GNT3*, *GALNT2*, *MUC3A*, *GCNT3*, *MUC5B*, *MUC5AC*
Compartments	GOCC:0005794	Golgi apparatus	8	4.98E-08	*GALNT14*, *B3GNT3*, *GALNT2*, *MUC3A*, *GCNT3*, *MUC16*, *MUC5B*, *MUC5AC*
	GOCC:0005796	Golgi lumen	4	5.75E-05	*MUC3A*, *MUC16*, *MUC5B*, *MUC5AC*
	GOCC:0070701	Mucus layer	2	0.0066	*MUC5B*, *MUC5AC*

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3.4 | O-linked glycosylation patterns and associated glycans of MUC5B

The functional significance of O-linked glycosylation for the immunosuppression role of MUC5B has encouraged us to check and analyze the probable glycosylation sites in MUC5B. This is important because the variations in glycosylated MUC5B can be a significant factor while designing cancer therapeutics against MUC5B for LUAD. O-linked glycosylation sites were predicted resulting in possible 1484 serine/threonine (Ser/Thr) sites across the protein. Five tandem repeats rich in serine and threonine amino acids were identified across the protein, namely 1890–2199, 2419–2756, 2976–3456, and 4233–4879. Among them, 150 O-linked glycosylation sites were predicted in 1890–2199 repeat, 169 sites in 2419–2756 repeat, 239 sites in 2976–3456 repeat, 171 sites in 3676–4013 repeat and 334 sites in 4233–4879 repeat (Table S4A). A total of 27 glycans were associated with MUC5B protein and is provided in Table S4B, however, the site of attachment of the identified glycans was not reported.

3.5 | Gene regulatory network of MUC5B

To study the co-relation between glycosylation and tumor promotion through RTKs, we constructed a regulatory network of MUC5B, EGFR, a well-known RTK of LUAD, and the β-N-acetylglucosaminyltransferase (OGT). The regulatory network of *MUC5B*, *EGFR*, and *OGT* identified the regulatory elements such as transcription factors and miRNAs modulating the expression of the genes. Transcription factors *NFKB1*, *RELA*, and *STAT3* were the common transcription factors regulating *MUC5B* and *EGFR*. Further, the common miRNA regulating *MUC5B*, *EGFR*, and *OGT* were hsa-mir-155-5p and hsa-mir-182-5p (Figure 3A,B). The complete list of regulating factors of *MUC5B*, *EGFR*, and *OGT* are provided in Table S5.

3.6 | Association between MUC5B and immune cells in LUAD

Estimation of tumor-immune infiltration revealed mutant MUC5B having higher expression with T cell CD8+ (log2FC: 0.587), T cell CD4+ (log2FC: 0.637), B cell (log2FC: 0.218), macrophage (log2FC: 0.532), cancer-associated fibroblasts (log2FC: 0.308) and myeloid-derived suppressor cells (MDSC) (log2FC: 2.233) as depicted in Figure 4A–F. Association of MUC5B expression and mutant MUC5B with immune inhibitors and stimulators identified increased expression of immune inhibitors such as CTLA4, IL10, PDCD1, PDCD1LG2, TIGIT, and LAG3 with mutant MUC5B as shown in Figure 5A–F. Furthermore, immuno-stimulators namely RAET1E, MICB, TMEM173, TNSFS13, TNSFS14, and TNSFS15 showed higher expression with mutant MUC5B while depicting negative correlation with MUC5B expression in LUAD. The correlation between MUC5B and immune-modulators is summarized in Table 2.

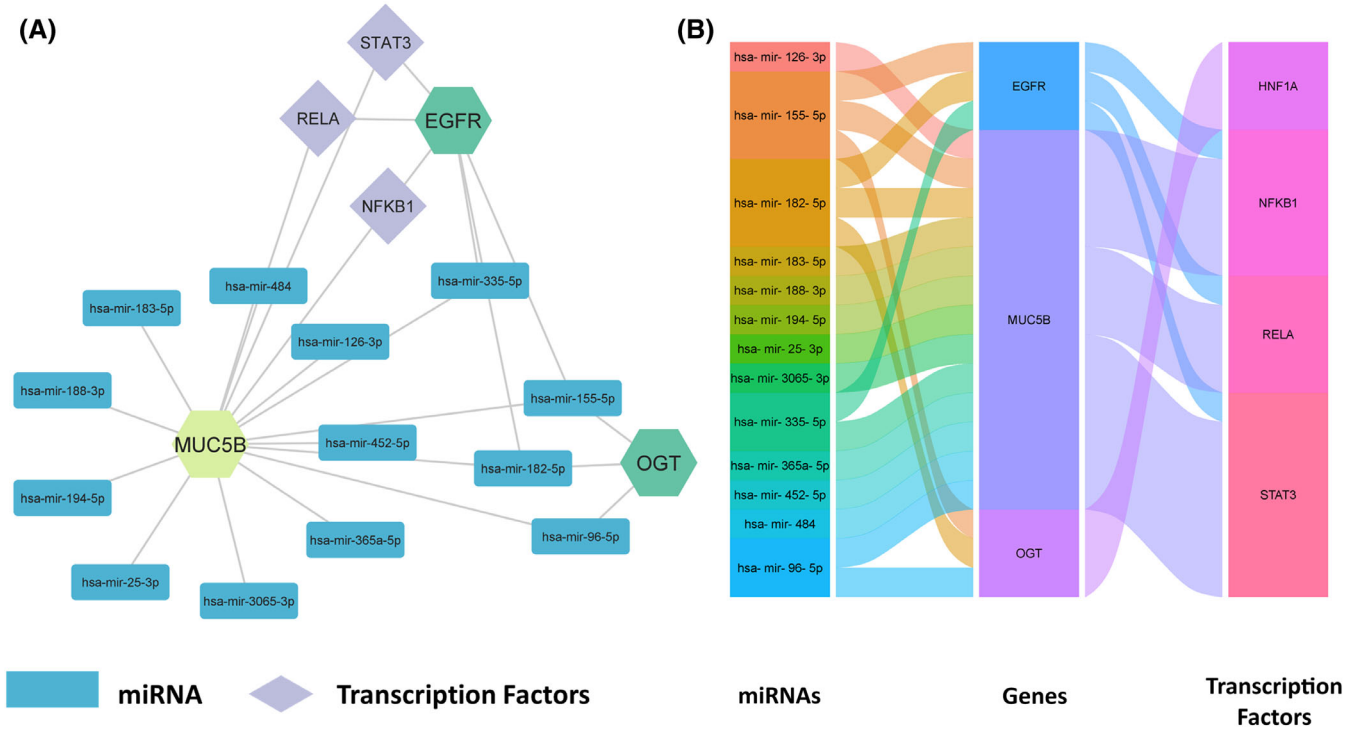


FIGURE 3 Gene regulatory network of *MUC5B*, *EGFR*, and *OGT* gene identified using miRNet server (A) Gene Regulatory network (B) Common miRNAs, Transcription factors among MUC5B, EGFR, and OGT.

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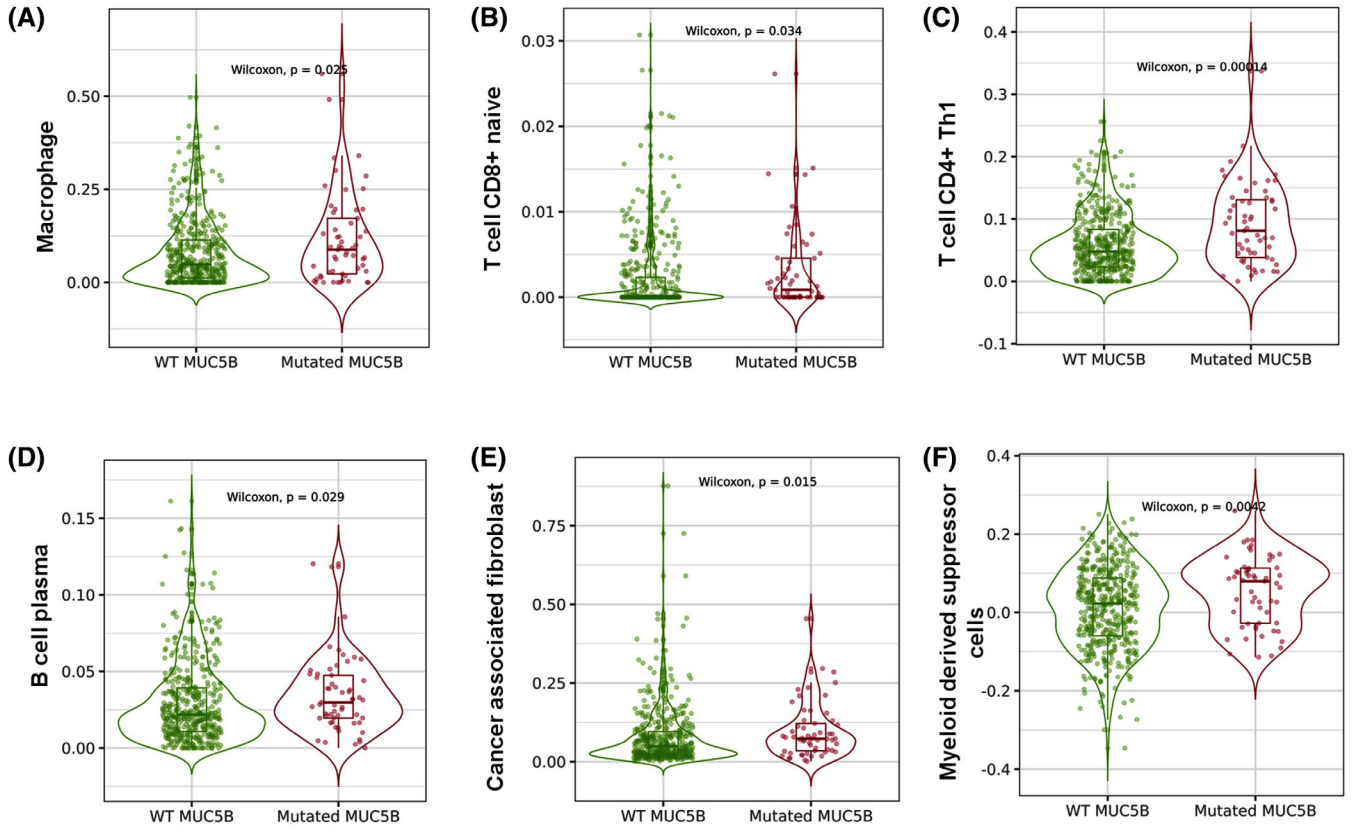


FIGURE 4 Estimation of immune cell infiltration with wild and mutant MUC5B (A) Macrophage (B) T cell CD8+ naïve (C) T cell CD4Th1 (D) B cell plasma (E) Cancer-associated fibroblasts (F) myeloid-derived suppressor cells.

3.7 | Genetic landscape of MUC5B and comparison with established biomarkers of LUAD

Mutational landscape of MUC5B gene in clinical samples of LUAD patients showed around 11% genetic alterations including amplification, deep deletion, and missense mutations in 2830 samples spanning across 10 studies namely LUAD Broad, Cell 2012 (183 samples); CPTAC, Cell 2020 (110 samples); MSKCC, 2020 (604 samples); MSKCC, 2021 (186 samples); MSKCC, Science 2015 (35 samples); NP J Precision Oncology, MSK 2021 (426 samples); OncoSG, Nat Genet 2020 (305 samples); TCGA Firehose Legacy (586 samples); TSP, Nature 2008 (163 samples) and NCI, Nature Genetics 2017 (232 samples). The clinically validated biomarker of LUAD namely, EGFR and KRAS showed a genetic alteration of 30% and 26% respectively as depicted in Figure S1.

3.8 | Functional and structural stability of mutant MUC5B

Among the 49 SNPs identified corresponding to the functional domains of MUC5B, 29 (59.18%) SNPs revealed neutral functional effect while 20 (40.81%) showed a deleterious effect (Figure 6A) as listed in Table 3. The structural stability identified seven SNPs namely N476Y (ΔΔG: 0.24 kcal/mol), S804I (ΔΔG: 0.31 kcal/mol), S912I (ΔΔG: 0.27 kcal/mol), H1418L (ΔΔG: 0.42 kcal/mol), A3621V (ΔΔG: 0.2 kcal/mol), A5300V (ΔΔG: 0.04 kcal/mol) and C5451F (ΔΔG: 0.05 kcal/mol) to be stabilizing mutations (Figure 6B). The stabilizing mutations were identified to be present in vWFD domain (N476Y), VWFC domain (S804I, S912I, A5300V, and C5451F), WxxW domain (H1418I and A3621V). Among the destabilizing mutations, 23 (54.76%) SNPs showed neutral effect on the protein function. However, the overall NMR chemical shift representing the relationship between the amino acids and backbone dynamics showed the wild MUC5B with an average S^2 of 0.755 while the mutant MUC5B with an average S^2 of 0.756 (Figure 6C).

3.9 | Comparative expression profiling of mucins in LUAD

The comparative gene expression profiling of MUC5B showed an average gene expression of 2.51 (logFC), which is higher than all other mucins for LUAD samples (Table S6) and the gene expression (expressed in logFC) of *MUC5B* across TCGA-LUAD study were found to be 3.78 with *p*-value of 1.366e-34 as represented in Figure 7A,B.

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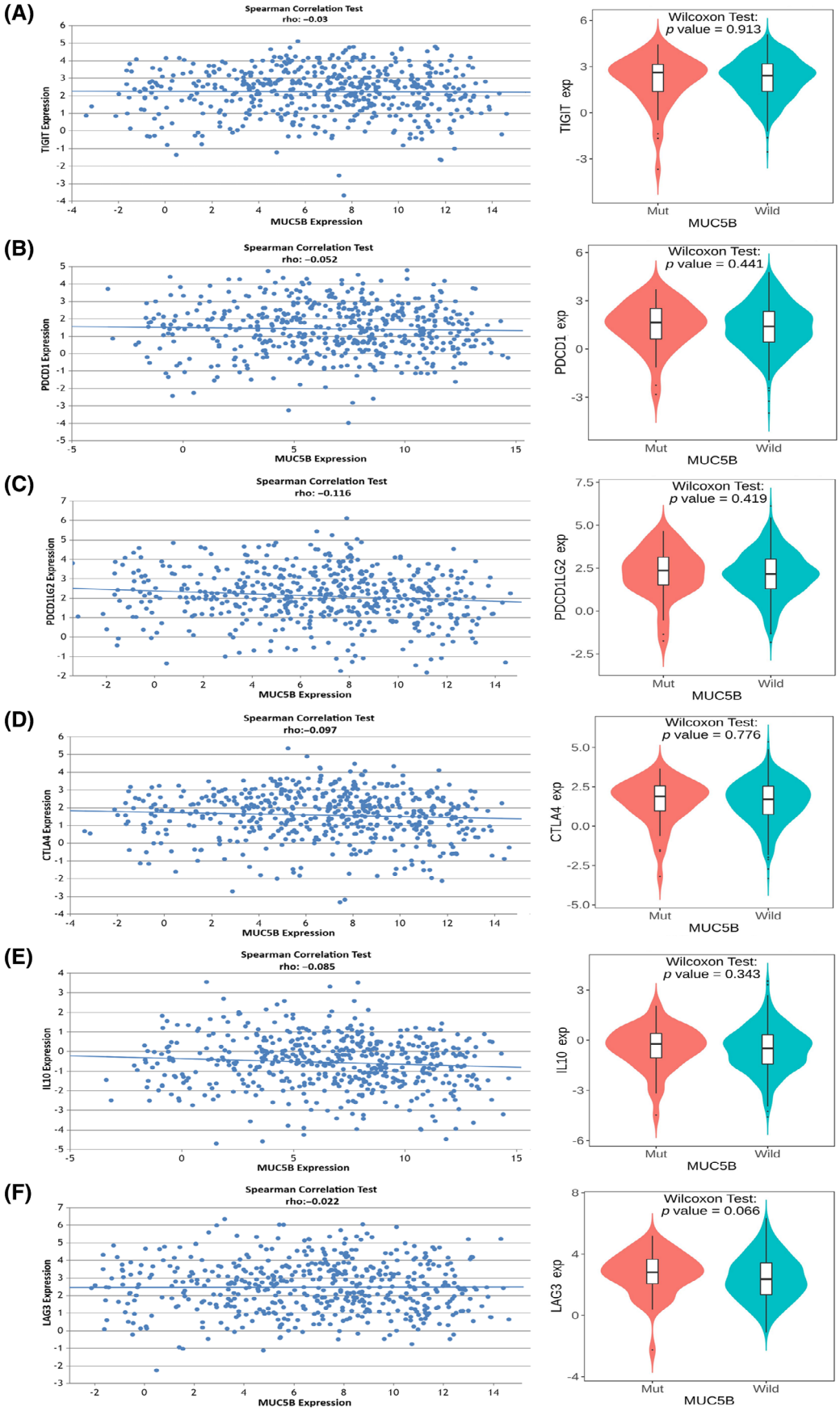


FIGURE 5 Correlation of immune inhibitors with MUC5B expression and mutation in lung adenocarcinoma (A) TIGIT (B) PDCD1 (C) PDCD1LG2 (D) CTLA4 (E) IL10 (F) LAG3. Correlation analysis was performed using Spearman correlation, while the relation between mutant and wild MUC5B with immune inhibitors was statistically performed using the Wilcoxon test.

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TABLE 2 Tumor-immune interaction analysing correlation between MUC5B and immunomodulators using the Spearman correlation and the Wilcoxon test.

Type of immune-modulators	Immunomodulators	Expression (Exp)	Mutation (Mut)
Immuno-inhibitors	CTLA4	−0.097	0.189
	IL10	−0.085	0.265
	PDCD1	−0.052	0.247
	PDCD1LG2	−0.116	0.203
	TIGIT	−0.03	0.199
	LAG3	−0.022	0.443
Immuno-stimulators	PVR	0.123	0.236
	RAET1E	0.191	−0.328
	CD86	−0.125	0.23
	MICB	−0.21	0.122
	TNFRSF25	−0.107	0.428
	TNFRSF9	−0.149	0.31
	TNFRSF4	−0.119	0.388
	ULBP1	−0.264	0.341
	TMEM173	0.46	−0.483
	TNFRSF13B	0.118	0.177
	TNFSF13	0.141	−0.47
	TNFSF14	0.227	−0.401
	TNFSF15	0.17	−0.486

3.10 | Diagnostic biomarker validation

The top five significant genes showing altered gene expression due to mutated MUC5B were MARC2 (LogFC: −1.64, *p*-value: 1.16E-06), ASPM (LogFC: 2.42, *p*-value: 2.2E-05), LOXLI-AS1 (LogFC: 1.49, *p*-value: 2.51E-05), NMU (LogFC: 1.62, *p*-value: 3.19E-05) and LINC00665 (LogFC: −1.45, *p*-value: 6.59E-05) as shown in Figure 7C–G. The deregulated genes with mean expression in wild and mutant are listed in Table 4.

3.11 | OA analysis

Kaplan Meier plot computed the survival outcome over time of MUC5B across a sample size of 239, with an HR value of 1.4 and *p*-value of 0.025. The overall MS identified the high expression of MUC5B showing lower percent survival (<50 months) when compared to the low expression of MUC5B. This coincides with the HR of 1.22 with a *p*-value of 0.026 identified using Km plotter. The MS of overall LUAD patients were identified to have a difference of ∼20% among the high and low groups. However, significant differences in the MS among the expression groups were identified for Stage 1 and Stage 3 samples. Stage 1 (with sample *n*: 907) showed an HR of 1.56 (*p*-value: 0.026) had an MS difference of ∼26%, where low group showed an MS of 72.33 months, while high expression group showed 46.2 months. Similarly, Stage 3 group with a sample size of 112, indicated a HR of 5.43 (*p*-value: 0.0015). However, an evident decrease of ∼68% in the MS was shown by the high expression group (MS_{high}: 13.33 months; MS_{low}: 81.1 months) as depicted in Figure 8A–C. Stage 2 samples did not yield a significant test result, and hence were not included in the study.

3.12 | Clinical profiling

The clinical profile of LUAD samples across different cohorts was then identified in pathological tumor (T), N, and M stages. Among the 499 samples of pathological T-stage LUAD samples, MUC5B gene had the highest median of 29.3 at T3 stage, while in 498 N samples, MUC5B had a median of 29.3 at N2 stage. MUC5B gene in 395 metastatic LUAD samples showed the highest median of 15.8 corresponding to the M1 stage. However, to validate our results statistically, we have investigated the median expression of MUC5B across tumor and N stages using the UALCAN database. When compared to the normal samples (ME: 0.587), stage 1 showed a median expression of 4.362 (*p*-value: 2.34e-08), while stage 2, stage 3, and stage 4 showed a ME of 13.748 (*p*-value: 3.35e-08), 13.544 (*p*-value: 4.06e-05) and 10.468 (*p*-value: 1.03e-02) as depicted in Figure 8D. Similarly, for the nodal M, statistically significant ME was high at the N2 stage (ME: 17.043; *p*-value: 3.54e-05) as shown in Figure 8E. We have summarized the rationale and outcomes of each analysis in Table 5.

4 | DISCUSSION

LUAD is unique in terms of aberrant expression of mucins that is clinically associated with cancer development and metastasis. The selective expression of mucins in LUAD makes them a critical factor for

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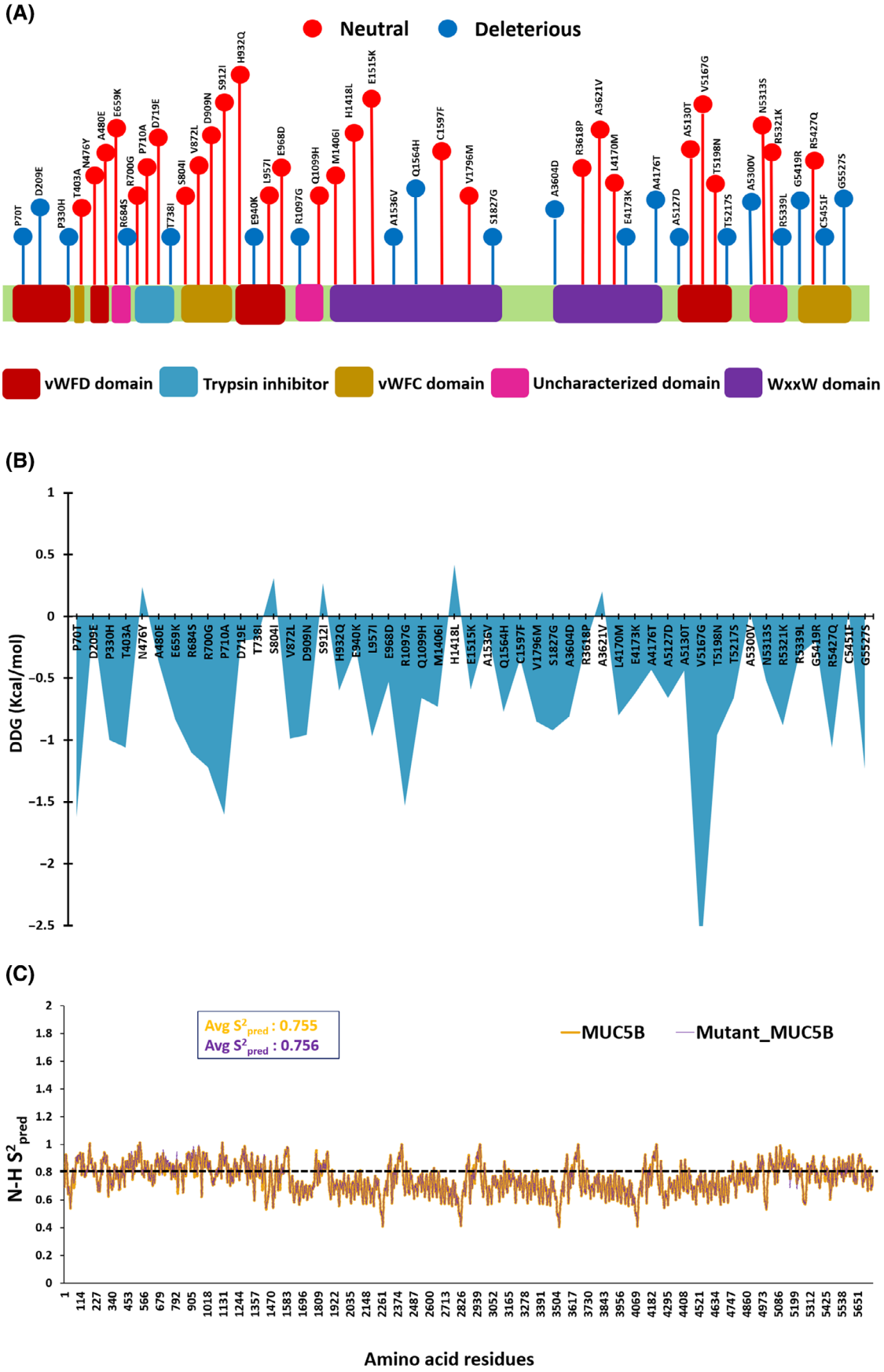


FIGURE 6 Functional and structural characterization of single nucleotide polymorphisms (SNPs) in MUC5B (A) Neutral and deleterious SNPs across functional domains of MUC5B (B) Stabilizing and destabilizing mutations in MUC5B (C) Backbone dynamics depicting N–H bond order using S^2 pred value in wild and mutant MUC5B. DDG represents the change in unfolding free energy.

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TABLE 3 Functional effect of single nucleotide polymorphisms associated with functional domains of MUC5B.

Domain (InterPro ID)	Amino acid substitution	Frequency	Functional effect	Accuracy	ΔΔG (kcal/mol)	Stability
Trypsin inhibitor-like, cysteine rich domain [IPR002919]	P300H	1	Deleterious	0.755	−1.00	Decrease
	R700G	4	Neutral	0.625	−1.22	Decrease
	P710A	2	Neutral	0.631	−1.60	Decrease
	D719E	1	Neutral	0.736	−0.20	Increase
	T738I	2	Deleterious	0.635	−0.18	Increase
Uncharacterised domain, cysteine rich [IPR014853]	E659K	1	Neutral	0.628	−0.83	Decrease
	R684S	2	Deleterious	0.718	−1.10	Decrease
	R1097G	4	Deleterious	0.643	−1.53	Decrease
	Q1099H	2	Neutral	0.826	−0.66	Decrease
	A5300V	2	Deleterious	0.643	0.04	Increase
	N5313S	2	Neutral	0.826	−0.52	Decrease
	R5321K	4	Neutral	0.738	−0.88	Decrease
	R5339L	1	Deleterious	0.505	−0.31	Decrease
von Willebrand factor type C [VWFC] domain [IPR001007]	T403A	2	Neutral	0.826	−1.06	Decrease
	S804I	1	Neutral	0.602	0.31	Increase
	V872L	1	Neutral	0.826	−0.99	Decrease
	D909N	1	Neutral	0.826	−0.96	Decrease
	S912I	2	Neutral	0.614	0.27	Increase
	G5419R	1	Deleterious	0.654	−0.20	Decrease
	R5427Q	2	Neutral	0.736	−1.06	Decrease
	C5451F	2	Deleterious	0.869	0.05	Increase
	G5527S	2	Deleterious	0.505	−1.23	Decrease
von Willebrand factor type D [VWFD] domain [IPR001846]	P70T	6	Deleterious	0.718	−1.62	Decrease
	D209E	2	Deleterious	0.654	−0.05	Increase
	N476Y	1	Neutral	0.602	0.24	Increase
	A480E	2	Neutral	0.602	−0.36	Decrease
	H932Q	2	Neutral	0.826	−0.60	Decrease
	E940K	3	Deleterious	0.521	−0.25	Decrease
	L957I	6	Neutral	0.826	−0.97	Decrease
	E968D	2	Neutral	0.826	−0.53	Decrease
	A5127D	1	Deleterious	0.505	−0.66	Decrease
	A5130T	1	Neutral	0.752	−0.44	Decrease
	V5167G	1	Deleterious	0.869	−2.73	Decrease
	T5198N	2	Neutral	0.602	−0.96	Decrease
	T5217S	1	Deleterious	0.605	−0.66	Decrease
WxxW [IPR025155]	M1406I	1	Neutral	0.752	−0.73	Decrease
	H1418L	2	Neutral	0.747	0.42	Decrease
	E1515K	1	Neutral	0.614	−0.59	Decrease
	A1536V	3	Neutral	0.736	−0.10	Decrease
	Q1564H	1	Deleterious	0.869	−0.77	Decrease
	C1597F	3	Deleterious	0.869	−0.34	Decrease
	V1796M	4	Neutral	0.625	−0.85	Decrease
	S1827G	4	Neutral	0.826	−0.92	Decrease
	A3604D	1	Deleterious	0.635	−0.81	Decrease
	R3618P	4	Deleterious	0.869	−0.23	Decrease
	A3621V	3	Neutral	0.738	0.20	Increase

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TABLE 3 (Continued)

Domain (InterPro ID)	Amino acid substitution	Frequency	Functional effect	Accuracy	ΔΔG (kcal/mol)	Stability
	L4170M	1	Neutral	0.738	−0.80	
	E4173K	1	Deleterious	0.521	−0.62	Decrease
	A4176T	3	Deleterious	0.505	−0.43	Decrease

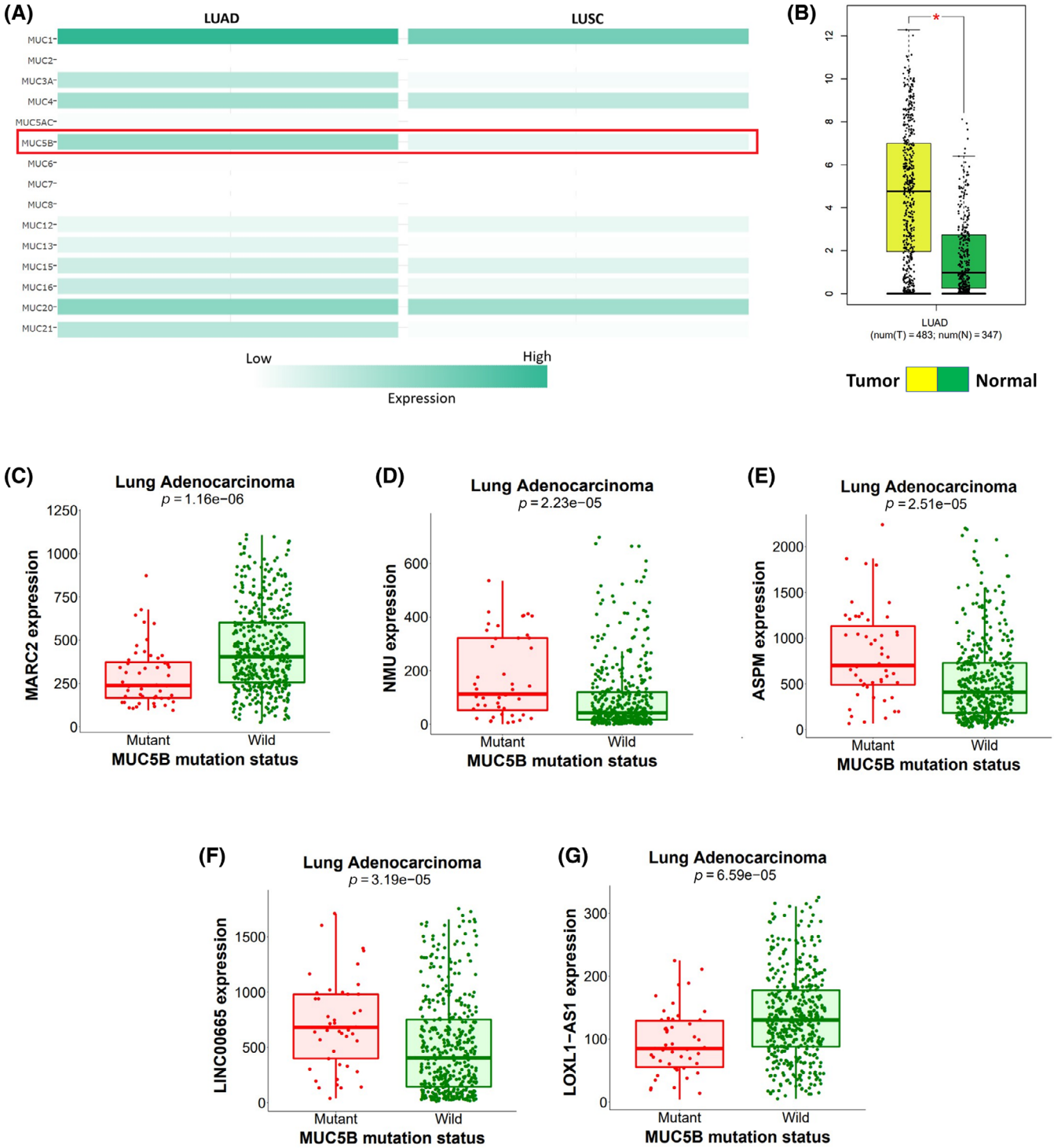


FIGURE 7 MUC5B biomarker validation (A) Comparative expression profiling of mucins across lung adenocarcinoma (LUAD) and lung squamous cell carcinoma (LUSC) (B) Boxplot depicting differential expression of MUC5B in LUAD (C–G) Significant genes with altered expression due to mutant MUC5B.

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TABLE 4 Top five significantly deregulated genes due to mutant MUC5B gene in lung adenocarcinoma.

Genotype					
Expression	**Mean mutant ($n = 52$)**	**Mean wild ($n = 456$)**	**LogFC**	**Direction**	***p*-Value**
MARC2	292.85	482.08	1.64	Down	1.16E-06
NMU	324.75	134.31	2.42	Up	2.23E-05
ASPM	854.33	575.14	1.49	Up	2.51E-05
LINC00665	977.81	603.76	1.62	Up	3.19E-05
LOXL1-AS1	102.98	149.27	1.45	Down	6.59E-05

clinical intervention. Deregulated MUC5B expression was reported in LUAD, but its mechanism of action or the role of functional interactors is not yet clearly understood. Here, we studied the molecular role of MUC5B in lung cancer pathogenesis, which identified the mutant and O-linked glycosylated MUC5B to significantly promote tumor proliferation by immune suppression mediated by glycosylation and TME components. This was supported by the enrichment of o-linked glycosylation and regulatory network. The correlation of mutant MUC5B with immune inhibitors further necessitated the need to categorize the nsSNPs of MUC5B. Lastly, the theranostic potential of MUC5B was validated using comparative expression, OA, and clinical profiling.

The deregulated *GALNT12* governs the regulation of mucin O-linked glycosylation and has been reported in colorectal cancer.[46] The increased glycosylated mucins interact with sialic acid binding immunoglobin-like lectins, siglecs that are expressed on natural killer cells, macrophages, and monocytes. This interaction results in the release of several tumor-promoting factors and hypersialyation-induced immune suppression. The oncogenic immune suppression results in the immune system being unable to recognize the self-cells from non-self-cells, which are otherwise recognized using sialic acid patterns on the cell surface under normal conditions.[47] Therefore, the enrichment of *MUC5B* gene in defective GALNT12 causing colorectal cancer (HSA-5083636), O-linked glycosylation of mucins (HSA-913709), immune systems (GO:0002376), respiratory mucus (BTO:0003603) and tracheobronchial epithelium (BTO:002925) suggest the possible mechanism of action of MUC5B in LUAD.

Majorly five tandem repeats rich in serine/threonine were found in MUC5B protein that have stable O-linked glycosylation. The differential glycosylation in target proteins leads to the formation of glycans that can aid in tumor proliferation. These tumor-associated glycans that are present on the cancer cell surface play crucial role in sustaining proliferation and TME-induced metastasis.[48] Moreover, glycosylation sites are critical as glycosylation present domains increase the mucin mass by several-fold change. The addition of glycosylation and subsequent formation of glycans to the mucin backbone influence the physical properties of the structural modifications. Hence, glycosylation pattern is a key determinant while designing therapeutics for MUC5B.[49–51] Therefore, detecting the glycosylation patterns in MUC5B and its associated glycans using antibody-mediated or mass spectrometry can be considered as a desirable theranostics for LUAD. Additionally, the altered glycosylation in mucins promotes cancer progression by modulating cell adhesion, migration, and immune evasion. This can be achieved due to the ability of mucins to alter the glycosylation of different RTKs leading to hyperactivation of RTKs, which in turn promotes RTK-dependent malignancy.[52] The interaction data of LUAD-associated genes showed the experimentally determined interaction between MUC5B and EGFR (CS: 0.41), a clinically proven RTK playing a major role in lung cancer. This suggests the possible role of MUC5B in promoting O-linked glycosylation in EGFR. The O-linked glycosylation catalyzed by β-N-acetylglucosaminyltransferase (OGT) enzyme was also seen to be significantly up-regulated across the four clinical stages in LUAD. The TFs namely, *NFKB1*, *RELA*, and *STAT3* were identified to be the common regulatory factors of *EGFR* and *MUC5B*. Furthermore, hsa-mir-155-5p and hsa-mir-182-5p, which were the common miRNAs regulating *MUC5B*, *EGFR*, and *OGT* play a major role in oncogenesis and have been associated with hypoxia-inducing and cell invasion in various tumors.[53,54] The common regulatory factors controlling these genes might suggest the interrelation between MUC5B-dependent glycosylation and EGFR-dependent tumor proliferation.

Tumor infiltrating lymphocytes (TILs) in the TME are associated with poor prognosis as the TME components sequentially change the tumor site, resulting in inflammation-induced cancer progression.[55,56] Here, the increased expression of immune cells due to mutant MUC5B suggests the role of immune cells in promoting tumor proliferation. The increased expression of CAFs (stromal component of TME) and MDSC with mutant MUC5B further supports the role of immune cells associated with TME in LUAD.[57] Immunomodulators including immune inhibitors and stimulators are predictive factors for treatment. Continued over-expression of immuno-inhibitors such as IL-10, CTLA4, TIGIT, and LAG3 were associated with enhanced cancer progression.[58–60] Among the immune-inhibitors, the correlation of mutant MUC5B to LAG3 can be significantly explored. This is because the lymphocyte activation gene 3 is required for T cell regulation and homeostasis. The continuous inflammation induced by MUC5B leads to the overexpression of LAG3, which can result in the exhaustion of T cells.[61] Furthermore, blocking LAG3 suggests anti-tumor responses as overexpression of LAG3 was shown to down-regulate the expression of T cells among other immunomodulatory functions.[62] Similarly, a positive correlation of mutant MUC5B with immuno-stimulators such as PVR, TNFRSF family genes indicates the ability of MUC5B protein to elicit immune response as a result emphasizing its potential as immunotherapy-based therapeutics.

Moreover, the 11% genetic alterations in LUAD cancer patients were identified to be comparatively lesser than the clinically used

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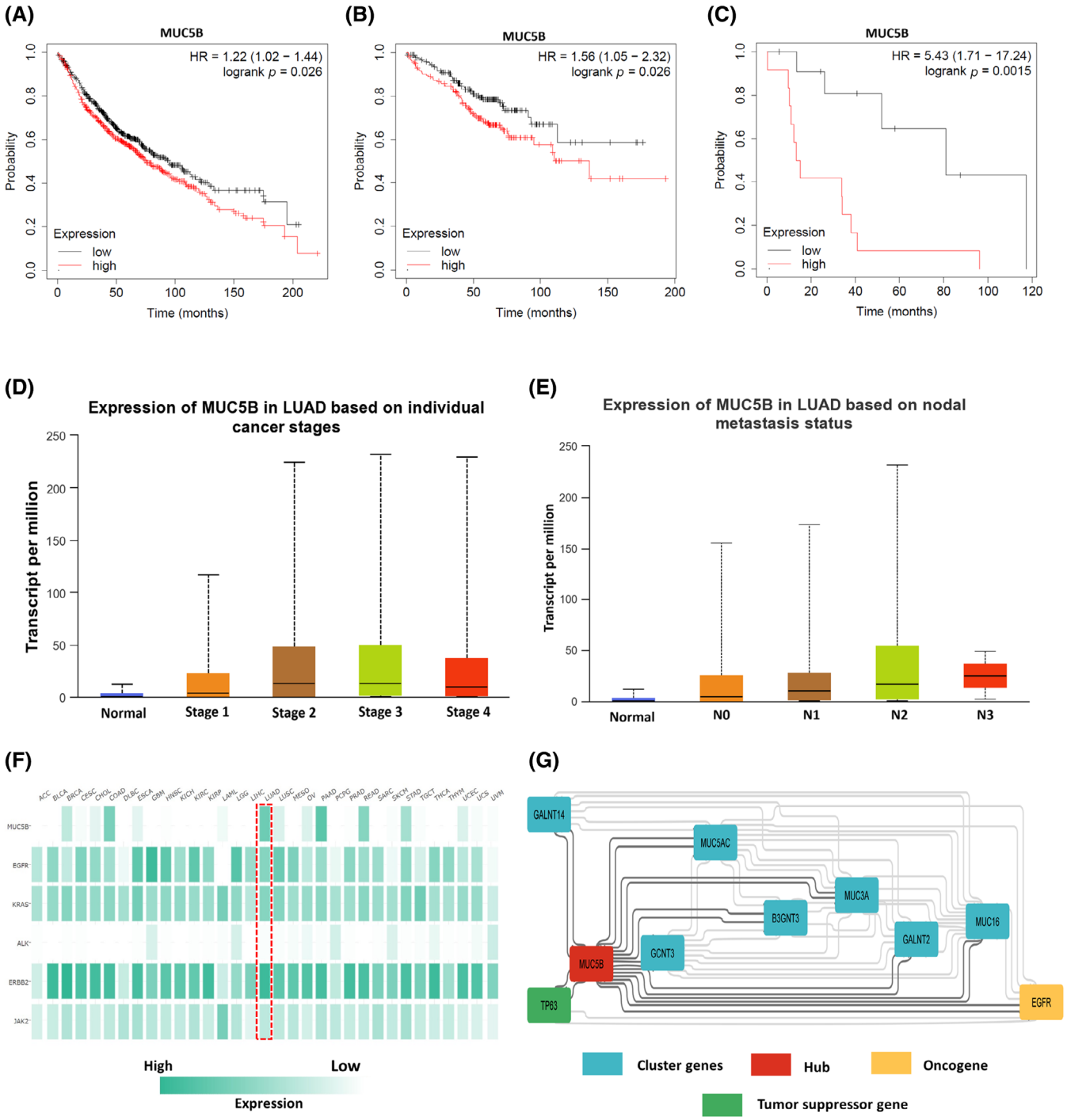


FIGURE 8 Survival and clinical profile analyses (A) Km plot representing overall survival of MUC5B in lung adenocarcinoma (LUAD); (B) Km plot representing overall survival of MUC5B in Stage 1 LUAD; (C) Km plot representing overall survival of MUC5B in Stage 3 LUAD; (D) clinical profile of the MUC5B at pathological tumor (T) stages; (E) clinical profile of MUC5B at lymph node (N) stages; (F) comparison of MUC5B expression with clinically validated biomarkers of LUAD; (G) direct interactors of MUC5B gene identified using LUAD interactome. HR, hazard ratio.

biomarkers such as EGFR and KRAS. However, the functional effect of the SNPs showed neutral and deleterious effect, which was then characterized by stabilizing and destabilizing mutations. The overall increase in the backbone N–H order represented by S^2 pred value by 0.1% in mutant MUC5B compared to parent MUC5B indicates the protein's rigidity for binding to macromolecules and continuing its molecular function. Moreover, the overexpression of the protein even in the presence of mutation indicates the retainment of overall function of MUC5B in LUAD, thus facilitating tumor progression.

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TABLE 5 Summarizing the major findings of each interactome and validation studies of MUC5B.

Interactome analysis	
1. Differential gene expression analysis	
Rationale: To investigate the expression profile of MUC5B	*Outcomes*: MUC5B gene was identified to be significantly up-regulated with a logFC of 2.36.
2. Interaction data curation and Topological parameter analysis	
Rationale: To study the direct interactors and influence of the over-expressed MUC5B gene in LUAD interactome	*Outcomes*: MUC5B was identified to be clustered along with closely associated genes such as *GCNT3*, *MUC5AC*, *B3GNT3*, *GALNT14*, *MUC3A*, *MUC16*, and *GALNT2*.
3. Functional enrichment analysis	
Rationale: To identify the functional role of MUC5B-present cluster genes in LUAD	*Outcomes*: Enriched pathway highlighted the role of O-linked glycosylation of mucins and immune system process.
4. O-linked Glycosylation pattern and associated Glycans	
Rationale: To study the different patterns of glycosylation and glycans associated that can influence the immune therapeutics designed for MUC5B	*Outcomes*: Five tandem repeats rich in Serine/Threonine residues were seen to be highly susceptible for o-linked glycosylation. These glycosylation patterns can influence the overall physical property of MUC5B by altering the structural modifications.
5. Gene regulatory network	
Rationale: To study the correlation among mucins, RTKs and o-linked glycosylation	*Outcomes*: MUC5B, EGFR (RTK), and OGT, enzyme responsible for o-linked glycosylation were seen to be sharing a close regulatory role. Common miRNAs regulating the transcription of MUC5B, EGFR, and OGT were elucidated. The common transcription factors suggest a close regulation among MUC5B, EGFR and its glycosylation.
6. Correlation of MUC5B with immune cells	
Rationale: To validate the role of MUC5B in enriched immune system biological process by studying its correlation with immune cells	*Outcomes*: When compared to the MUC5B (wild), mutant MUC5B showed increased expression of immune inhibitor cells, indicating a probable anti-tumor response with the influence of mutant MUC5B.
7. Mutational profiling	
Rationale: To perform mutational profiling of MUC5B and comparative analysis with well-established LUAD biomarkers (EGFR and KRAS)	*Outcomes*: Comparative low mutational profile of MUC5B indicates its ability to be a therapeutic target for immunotherapy for LUAD.
8. Characterization of nsSNPs in MUC5B	
Rationale: To characterize the non-synonymous mutations in MUC5B	*Outcomes*: The nsSNPs were categorized into deleterious or neutral based on the functional phenotype while based on the structural stability the SNPs were categorized into increasing or decreasing stability.
9. Comparative gene expression	
Rationale: To investigate the gene expression of MUC5B across lung adenocarcinoma samples to confirm its potential as theranostic biomarkers	*Outcomes*: MUC5B was identified to be evidently and significantly up-regulated across lung adenocarcinoma tumor samples indicating its potential as biomarker.
10. Cancer biomarker validation	
Rationale: To explore the influence of the mutant MUC5B on LUAD genes for tumor progression	*Outcomes*: Significant and correlated genes were identified that were shown to promote tumor progression.
11. Overall survival analysis	
Rationale: To study the influence of MUC5B gene on the overall survival of lung adenocarcinoma patients	*Outcomes*: MUC5B gene showed an evident decrease in the median overall survival at tumor stage 3.
12. Clinical profiling	
Rationale: To perform clinical profiling of the MUC5B gene to identify the tumor and nodal metastasis-specific stages and its median expression	*Outcomes*: Increased median expression was identified at Tumor stage 3 as well as in nodal metastasis stage 2. This coincides with the overall survival plot indicating an increased expression of MUC5B results in decreased overall survival.

Abbreviations: LUAD, lung adenocarcinoma; RTKs, receptor tyrosine kinases; SNPs, single nucleotide polymorphisms.

The functional significance of *MUC5B* gene in tumor progression was further indicated by its capability of deregulating tumor-promoting and suppressing genes. The mitochondrial amidoxime-reducing component 2 (*MARC2*) that controls cancer mobility by regulating p27 is down-regulated due to mutated *MUC5B*. Increased p27 under normal conditions causes cell cycle arrest,[63]

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hence down-regulation of *MARC2* due to mutant MUC5B causes cell proliferation in LUAD. Similarly, up-regulation of Neuromedin U (*NMU*) that plays a key role in immune-mediated inflammation, and promotes tumor growth in LUAD as a result of mutant *MUC5B*.[64] Additionally, the abnormal spindle-like microcephaly-associated protein coding gene *ASPM* is up-regulated due to mutated *MUC5B*. Knockdown of the *ASPM* gene has demonstrated reduced bladder cancer cell proliferation[65] indicating the possible mechanism of oncogenesis. The long non-coding RNA, LINC00665 regulated by transcription factor *SP1* exerts an oncogenic role by activating ERK signaling and its up-regulation by mutant *MUC5B* in LUAD can be a possible pathway aiding in tumor progression.[66] Mutant *MUC5B* was seen to down-regulating *LOXL1-AS1*, thus preventing the NF-κB pathway activation.[67] The deregulation of tumor-promoting genes as a consequence of mutated *MUC5B* indicates its critical role in promoting lung cancer.

We further analyzed the specificity of *MUC5B* as a potential diagnostic biomarker when compared across other clinically validated biomarkers of LUAD (Figure 7F). The differential expression of the *MUC5B* gene was seen to be specifically expressed in LUAD and not in lung squamous cell carcinoma (LUSC), further proving MUC5B specificity. The OA analysis indicated the high expression of MUC5B with reduced MS when compared to low expression, indicating the poor survival outcome of MUC5B in LUAD. This was evident in the case of the MS in stage 3 for LUAD samples. Furthermore, the survival outcome analysis coincides with the expression patterns across the tumor stages (T3) and N M (N2). The increased expression of *MUC5B* in advanced cancer stages clearly indicates its potential to be a translatable diagnostic biomarker. *MUC5B* was also seen to be interacting with a clinically validated biomarkers of lung cancer such as *EGFR* oncogene (CS: 0.411) and tumor suppressor gene (TSG) *TP63* (CS: 0.517) as represented in (Figure 7G). Variations in the *TP63* gene have been reported to be influencing patients' susceptibility towards LUAD.

The current study has used extensively in silico methods to characterize the MUC5B gene's role in LUAD. Considering the limitations of in silico methodologies, we have validated our results using clinical samples of more than 600 cancer patients from TCGA and GTEx Database.[68] It has been reported that big data analysis by integrating with a global consortium comprising experimentally characterized regulatory molecules can potentiate the recommendation of a biomarker by circumventing the bias of small/regional sample size. To justify the sample size and diversity the datasets used are outlined as follows: The clinical data obtained from RNA sequencing is processed and normalized and expressed in terms of TPM. Also, the mutational profiling of MUC5B was conducted by a dataset consisting of 2830 samples spanning across 10 studies namely, LUAD Broad, Cell 2012 (183 samples); CPTAC, Cell 2020 (110 samples); MSKCC, 2020 (604 samples); MSKCC, 2021 (186 samples); MSKCC, Science 2015 (35 samples); NP J Precision Oncology, MSK 2021 (426 samples); OncoSG, Nat Genet 2020 (305 samples); TCGA Firehose Legacy (586 samples); TSP, Nature 2008 (163 samples) and NCI, Nature Genetics 2017 (232 samples).[37] Likewise, the immune cell association of MUC5B in LUAD was analyzed with a sample size of 515 LUAD clinical samples.[35,36] We are endeavoring to create datasets, which will be useful for future machine learning-artificial intelligence-based biomarker predictions leading to diagnostic breakthroughs. Therefore, the direct interaction of the *MUC5B* gene with factors affecting LUAD progression further encourages us to consider *MUC5B* as a potent diagnostic biomarker for LUAD.

5 | CONCLUSION

Glycosylated mucins play a major role in lung tumor progression and metastasis, however, the role of *MUC5B* was not clearly understood. Here, LUAD interactome identified the functional role and associated interactors of *MUC5B* highlighting its role in immune evasion and O-linked glycosylation. The modulation of glycosylation in serine–threonine-rich tandem repeats and mutation-induced alterations of TME components (immune cells) suggest the possible diagnostic and therapeutic role of *MUC5B* for immunotherapy-based clinical intervention.

AUTHOR CONTRIBUTIONS

Conceptualization: Sudha Ramaiah and Anand Anbarasu. *Project administration and funding acquisition*: Sudha Ramaiah and Anand Anbarasu. *Methodology*: Sudha Ramaiah and Anand Anbarasu. *Supervision*: Sudha Ramaiah and Anand Anbarasu. *Formal analysis and investigation*: Gayathri Ashok and Abirami Soundararajan. *Writing-original draft preparation*: Gayathri Ashok. *Visualization and graphical interpretation*: Gayathri Ashok.

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CONFLICT OF INTEREST STATEMENT

The authors declare no potential conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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